

$$a\vec{U} + b\vec{V} = 0$$

$$a=0 \quad b=0 \quad \text{LI}$$

$$a \neq 0 \quad b \neq 0 \quad \text{LD}$$

Se única Solução
 $a, b = 0 \rightarrow \text{LI}$

Exemplv:

$$\vec{U} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \vec{V} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$a\vec{U} + b\vec{V} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$1: a1 + b0 = 0$$

$$2: a0 + b1 = 0$$

$$2: \begin{array}{rcl} a0 + b1 & = & 0 \\ -a0 & & -a0 \end{array}$$

$$b1 = 0 - a0$$

$$b1 = 0 - 0 - b0$$

$$b1 = 0 - b0$$

$$b = 0 - 0$$

$$\underline{b = 0}$$

$$1: \begin{array}{rcl} a1 + b0 & = & 0 \\ -b0 & & -b0 \end{array}$$

$$a1 = 0 - b0$$

$$a1 = 0 - 0 \times 0$$

$$a1 = 0 - 0$$

$$a1 = 0$$

$$\underline{a = 0}$$

Respostas: Linearmente Independente